

HAT-P-37b

R-2601

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_220910 · created 2026-08-28T00:33:56+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459832.6977 +/- 0.0088 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.049
Transit depth (Rp/Rs)^2	2.49 +/- 1.56 [%]
Orbital Inclination (inc)	85.76 +/- 2.41
Airmass coefficient 1 (a1)	1.1 +/- 0.73
Airmass coefficient 2 (a2)	-0.01 +/- 0.46
Scatter in the residuals of the lightcurve fit is	75.59 %
Best Comparison Star	#3 - [595, 474]
Optimal Aperture	2.41
Optimal Annulus	3.0
Transit Duration (day)	0.075 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.158 $\pm$ 0.049 vs 0.1394 (deviation 0.21 σ)
Duration fit vs published	0.075 d vs 0.0983 d (deviation 0.44 σ)
Mid-time O-C	-20.4 min
Depth SNR	1.8
Residual scatter	75.59 %

Light curve

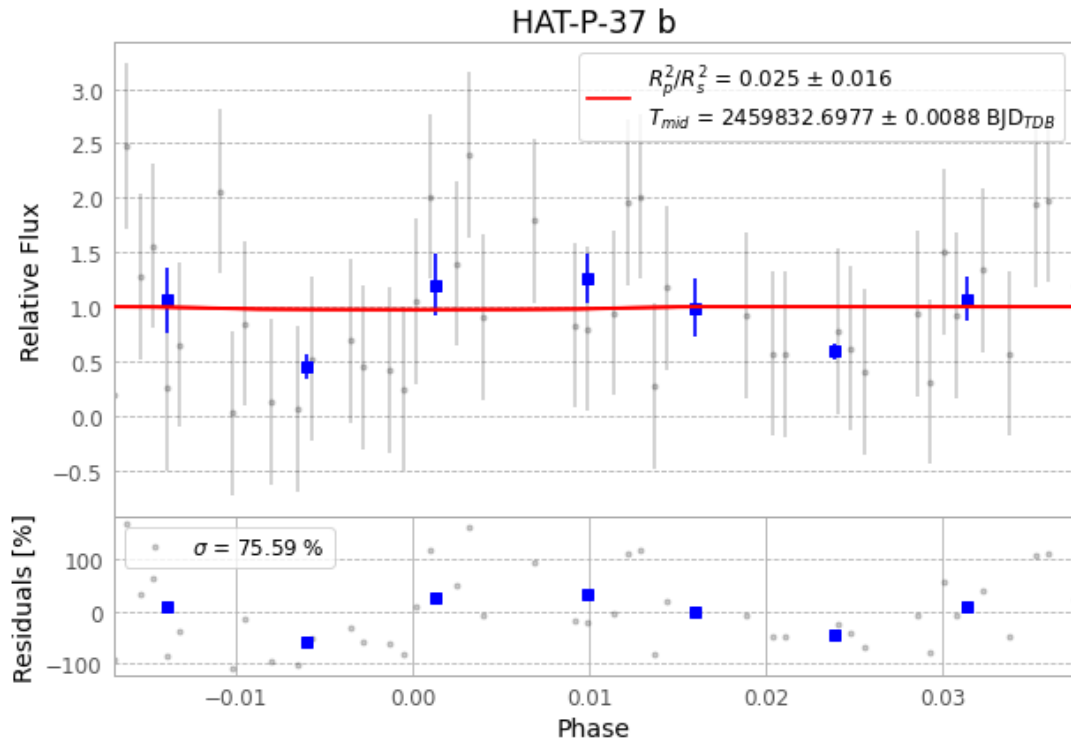


Figure 1 — FinalLightCurve_HAT-P-37 b_2022-09-09.png (SHA-256 68654a3ce5ea0956...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [430, 201], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e0cfc90040f82a9b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

75 FITS frames (49 MB), HATP-37220910033013.FITS → HATP-37220910071213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-220910.log (946570ca8edf...), FinalLightCurve_HAT-P-37 b_2022-09-09.png (68654a3ce5ea...), FinalLightCurve_HAT-P-37 b_2022-09-09.csv (d4707fd21524...)

Compute resources

Wall time 154.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 00:33Z.